INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
SIGNAL GENERATOR
1.8 GHZ TO 4.5 GHZ

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a 1.8 GHz to 4.5 GHz Signal Generator, hereinafter called the signal generator. The commercially available signal generator will be a laboratory standard, quality, bench piece of electronic test equipment that is capable of being rack-mounted, solid-state, signal generator.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and bid sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Order of Precedence. In the event of conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence. Nothing in this CID; however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2.3 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.3.1 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0º C to 50º C.

Comments, suggestions, or questions on this document should be addressed to: 542 ATSG/GBTRND 380 Richard Ray Blvd, Ste 104, Robins AFB, GA 31098. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

FSC 6625

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2.3.2 Non-operating Temperature. The equipment shall meet its performance and accuracy requirements after being subjected to a non-operating environment of -40º C to +71º C.

2.4 Electrical Power Sources. The signal generator shall operate from nominal commercial, military, and shipboard power sources of 110/220 volts AC (+/- 10%) at line frequencies of 45 Hz to 440 Hz.

2.5 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 4,000 hours of operation with a statistical certainty of 95%.

2.6 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

2.7 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions, and operating service condition as specified in MIL-PRF-28800.

2.7.1 Frequency Range. The signal generator shall have a frequency range of at least 1.8 GHz to 4.5 GHz.

2.7.2 Resolution. The signal generator shall have a resolution of 1 Hz or less.

2.7.3 Accuracy. The frequency accuracy shall be $1x10^{-9}$/day, $1x10^{-8}$/month or better.

2.7.4 RF Output. The signal generator shall have a minimum output range as follow:

1.8 GHz to 3 GHz : +10 dBm to -127 dBm
3 GHz to 4.5 GHz: +3 dBm to -127 dBm

2.7.5 Output Resolution. The signal generator shall have an output resolution of 1 dB or less.

2.7.6 Output Accuracy. The signal generator output accuracy shall be ±1 dB or better.

2.7.7 Spectral Purity.

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2.7.7.1 Harmonics. The signal generator shall have a harmonic of -25 dBc or better when the output is as follow:

1.8 GHz to 3 GHz: +7 dBm
3 GHz to 4.5 GHz: +3 dBm

2.7.7.2 Sub-Harmonics. The signal generator shall have a sub-harmonic as follows

≤ 2.3 GHz: -40 dBc or better
≤ 4.5 GHz: -30 dBc or better

2.7.7.3 Non-Harmonics. The signal generator shall have a non-harmonic of -70 dBc or better for offsets of 3kHz or greater.

2.8.8 Pulse Modulation (PM).

2.8.8.1 PM On/Off Ratio. On-Off ratio shall be at least 20 dB for square wave and pulse modulation.

2.8.8.2 PM Rise/Fall Times. The pulse modulation rise/fall time shall be 2 μs or less.

2.8.8.3 Internal Square Wave. The minimum frequency range shall be 950 to 1050 Hz.

2.8.9 Amplitude Modulation (AM). The AM mode of the signal generator shall have a depth of at least 0 to 80% at a resolution of 0.1%.

2.8.10 Frequency Modulation (FM).

2.8.10.1 FM Accuracy. The signal generator shall have an accuracy within ±5%.

2.8.10.2 FM Resolution. The signal generator shall have an FM resolution of at least 1 Hz or 1%.

2.8.10.3 FM Deviation. The signal generator shall have a peak deviation from 0 to 1 MHz for carrier frequencies above 1.8 GHz.

2.8.11 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a one-person lift of 37 pounds, (16.8 kilograms).

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2.8.12 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

2.8.13 Rack-mount. The analyzer shall be capable of being rack-mounted in a standard ANSI/EIA-310-D equipment rack and include a rack-mount conversion kit if needed.

2.9 Manuals. The equipment shall be delivered with operator, maintenance/service, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS

3.1 Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The product provided shall meet the salient characteristics of this purchase description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 Acceptance. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING

5.1 Packaging, Packing, and Marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Sources of documents.

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6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or from the Standardization Documents Order Desk, 700 Robbins Ave., Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.1.1 MIL-PRF-28800, Test Equipment for Use with Electrical and Electronic Equipment, General Specification for

6.1.2 The Code of Federal Regulations, (CFR), may be obtained online at http://www.gpoaccess.gov/cfr/index.html or from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 Institute of Electrical and Electronics Engineers (IEEE) standards products can be ordered online at http://www.ieee.org or the IEEE Operations Center. (Phone: 1-800-678-IEEE). Mail: IEEE Operations Center, Sales Office, 445 Hoes Lane, PO Box 1331, Piscataway, NJ, 08855-1331, USA.

6.1.3.1 IEEE Std 488.2-1992, IEEE Standard Codes, Formats, Protocols, and Common Commands for Use With IEEE Std 488.1-1987, IEEE Standard Digital Interface for Programmable Instrumentation.

6.1.4 Electronics Industry Alliance (ANSI/EIA). Hardcopy of EIA standards and technical publications are available online at http://global.ihs.com or from the Global Engineering Documents, the exclusive distributor of EIA standards. Phone number: 1-800-854-7179.

6.1.4.1 EIA-310, Cabinets, Racks, Panels, and Associated Equipment.

6.2 Ordering data. Procurement documents must specify the title, number, and date of this CID, the issue of the ASSIST Online, http://assist.daps.dla.mil to be used in the solicitation, and preservation and packaging requirements.

6.3 Key words.

Calibration
Generator
Signal

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MILITARY INTEREST

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

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